Pablo Borrego Ramos

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EDUCATION

ENS Paris-Saclay **Paris, France · Sep 2026 – Present**
MVA master's program (Mathématiques, Vision, Apprentissage)
Current coursework: Optimal Transport for ML, Stochastic Calculus in ML, Convex Optimization for ML, Distributed and Large-Scale Optimization, Statistical Learning, Geometric Data Analysis, Learning for Time Series, Deep Learning and Signal Processing.

Universidad de Extremadura **Badajoz, Spain · Sep 2020 – Jun 2025**
Dual Bachelor's Degrees in Mathematics & Statistics (5-year program)
Award for the Best Academic Record in the Double Degree; ranked 1st/50 in Statistics and 2nd/50 in Mathematics · Merit-based 99% tuition waiver (2021–2025).

Selected coursework: Functional Analysis, Measure & Integration Theory, Ordinary and Partial Differential Equations, Numerical and Computational Methods I–II, Operations Research (basic and advanced), Probability, Stochastic Processes (basic and advanced), Time Series, Multivariate Statistics.

Schools & conferences attended: XX DANCE Winter School in Dynamical Systems (RTNS 2024, Badajoz, 2024); Methods in Banach Spaces, satellite conference of the 9th European Congress of Mathematics (Badajoz, 2024).

RESEARCH

Research interests: optimal transport and its applications; stochastic processes and dynamical systems; mathematical and combinatorial optimization.

Preprints

An Exact Branch-and-Cut Method for the Asymmetric Orienteering Problem with Terrain-Driven Fatigue **2026**
Sole author. Under review at Computers & Operations Research. [code]
Introduces the Asymmetric Orienteering Problem with Fatigue (AOPF) over GIS-based terrain cost surfaces; develops two exact MILP formulations and a branch-and-cut solver with tailored cutting planes, including a novel family derived from the assignment relaxation. Certifies global optimality on 11 of 14 real-terrain instances, with the flow formulation reducing the mean gap from 20.0% to 11.4% relative to MTZ.

Theses

Discrete Dynamical Systems in Low Dimensions **2025**
BSc Thesis in Mathematics · Supervisor: Prof. José Luis Bravo Trinidad
Stability of non-hyperbolic fixed points of $\mathcal{C}^3$ maps (original criterion for $f'(\hat{x}) = \pm 1$); perturbations of monotone systems and period-doubling bifurcations; proof of chaos for the logistic family ($\lambda > 4$) via symbolic dynamics and conjugacy with the shift map.

Functional Data Analysis: Theory and Applications to Financial Time Series **2025**
BSc Thesis in Statistics · Supervisor: Prof. Julián Ramajo Hernández
L^2 framework for functional data: basis expansions, penalised smoothing, functional PCA (Karhunen–Loève); functional regression and clustering; application to 655 ECB yield curves (2006–2009) with the IBEX 35 as response ($R^2 = 0.81$, OLS), in R.

PROFESSIONAL EXPERIENCE

Amazon — *Business Intelligence Engineer (intern, then full-time)* **Luxembourg · Feb 2025 – Aug 2026**
- Statistical and ML models (penalised and Poisson regression, gradient boosting, ARIMA/ETS) for throughput prediction and capacity planning across 500+ warehouse machines.
- Designed an end-to-end absenteeism forecasting system for 50+ European sites with automated retraining, reducing forecast error (WAPE) by 30%.

Metanogenia — *Data Science Intern* **Spain · Jun – Jul 2023**
Methane-yield prediction from heterogeneous reactor data; benchmarked six regression methods with 5-fold CV ($R^2 = 0.71$).

ADDITIONAL INFORMATION

Programming: Python (PyTorch, NumPy, SciPy), R, C++, SageMath, Fortran, LaTeX. Numerical simulation, statistical modelling, exact optimization, scientific writing.
Tools: Git/GitHub, Linux.
Languages: Spanish (native), English (fluent, C1), French (Learning, A2).
Other: Faculty Council member (student representative), UEx; European Solidarity Corps volunteer (2024); ADMO marrow-donor awareness campaigns (2024–2025); Spanish Adventure Racing League Champion (2021).